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Highly-defective graphene as a metal-free catalyst for chemical vapor deposition growth of graphene glass

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ABSTRACT

Direct synthesis of graphene on insulating substrates is highly desirable for practical applications. Herein, we present a new fabrication method for graphene on quartz which enables the direct growth of high-quality and large-scale graphene films with no metal involved. The growth rate of graphene on insulating substrates was typically extremely slow due to the low catalytic activity and the slow diffusion speed of carbon species on them. Thus, an extra catalyst is required. Instead of metal elements, such as Cu, Ni, and Ga, highly-defective graphene films were applied as the catalyst for methane decomposition. The key feature of our approach is that carbon itself was used as the catalyst for graphene fabrication, which completely avoided the contamination of other elements. As a result, the growth rate of graphene films increased significantly compared to the conventional CVD method. The excellent quality of graphene glass was verified by a series of experimental characterization and the fabrication of a transparent heating device without any transfer process. In addition, the energy barrier of the decomposition of methane was explored using the first principles calculation, which further confirmed the catalytic behavior of highly-defective graphene films.

1. Introduction

Graphene, a two-dimensional material, has attracted massive attention due to its unique physical properties and various potential applications [1–4]. Given the requirement of high-quality and large scale of graphene films, great progress has been made in the development of novel synthesis techniques of graphene in the past few years. Among all the approaches, chemical vapor deposition (CVD) is thought to be the most promising [5,6]. CVD on metal foils such as Cu, Ni, Pt, Cu/Ni alloy has gained enormous success, which is capable of providing graphene films with desired quality, uniformity and scale [7–9]. Nevertheless, the transfer process of graphene films from the catalytic metal substrates to the insulating substrates is quite complex, and often introduces defects and contamination, which is a major issue for practical applications [10,11]. In addition to optimizing the transfer methods, developing CVD methods for direct growth of graphene on insulating substrates is another potential route.

To date, many efforts have been made toward direct synthesis for graphene on various insulating substrates, such as quartz, [12–15] sapphire, [16–18] SiC, [19,20] SiO_2, [21,22], Si_3N_4 [23] and so on. In particular, graphene grown on quartz has gained extensive attention due to its potential performance when applied as transparent heating devices, transparent electronic devices, defogging glass and touch panels [24]. However, the low catalytic activity of the quartz surface and slow diffusion speed of carbon species on the insulating substrates limit the growth rate and quality of graphene films. Introducing extra catalysts seem to be a natural solution. In view of the excellent catalytic performance of metal, some strategies have been attempted to introduce metal catalyst during direct CVD growth of graphene on insulating substrates. For example, by depositing a metal layer on quartz, a downward segregation of carbon atoms occurs at the metal/quartz interface, leading to the formation of graphene on quartz. This method can improve the growth quality and efficiency, but requires complex processes before and after CVD growth, the sacrifice of metal layers is wasteful as well [10,25,26]. More recently, metal vapors, metal

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nanoparticles or metal implantation were also employed in CVD methods as catalyst, which were proved to be remarkably efficient [14,17,27,28]. However, these approaches always remain a technical challenge in removing the metal contamination completely, which is essential for further applications. Thus, a metal-free catalytic method for graphene growth on insulating substrates is highly demanded.

In this work, we developed a metal-free CVD process to achieve a significantly increased graphene growth rate, involving highly-defective graphene films as catalyst (DG-CVD). With the assist of graphene defects and a narrow gap between catalytic plate and target substrate, the rate of methane cracking was highly improved. As a matter of fact, high-quality and single-layered graphene was synthesized in much shorter time compared to conventional CVD method. Moreover, by the combination of first principles calculation, the mechanism of this technique was investigated. The highlight of DG-CVD method was the use of graphene itself as the catalyst of graphene growth, which completely avoided the introduction of other elements.

2. Results and discussion

Fig. 1a and Fig. 1b show a schematic illustration of the DG-CVD approach. The highly-defective graphene glass (DG-Plate) was synthesized and employed as a catalyst for CVD growth of graphene on quartz substrates. The quartz substrate was placed above the DG-Plate. In high temperature condition, graphene defects on DG-Plate became catalytic sites for the decomposition of methane, leading to a higher concentration of active carbon radicals in the gas phase [29,30]. Meanwhile, DG-Plate was not only a catalyst, but also an additional carbon source [31]. Both effects resulted in an increase in the nucleation and growth rate of graphene. Thus, high density of defects on DG-Plate was the critical point in graphene growth of our method, which could be reflected by Raman spectrum (Fig. 1c and Fig. S7).

In our experimental design, a narrow gap was involved in DG-CVD method, which has a distance of ~5 μm (Fig. 1d). Initially, the gap was able to enhance the growth rate of graphene due to the trapping effect according to previous reports [7,13]. Methane molecules were trapped in the gap and the collision probability was significantly increased. In addition, the narrow gap was thin enough for active carbon radicals travel from catalytic plate (DG-Plate) to the surface of growth substrate. It was found that the catalytic effect of DG-Plate reduced dramatically with a larger distance of the gap.

To investigate the role of DG catalytic plate and narrow gap during the growth of graphene, a direct comparison experiment was performed using both DG-CVD and conventional CVD methods under the same synthesis conditions ($Ar/H_2/CH_4 = 200/45/2$ sccm, 2 h, 1130 °C & 1170 °C). It was worth noting that a narrow gap was also involved in conventional CVD process by replacing the DG-Plate with a pristine quartz plate as shown in Fig. 1b. The comparison of front side and back side of the conventional CVD sample (marked as 'w/o Catalyst-F' and 'w/o Catalyst-B' respectively) were able to testify the necessity of narrow gap naturally. Raman spectroscopy was used to characterize the number of layers and density of defects [32]. Although all of the samples displayed the characteristic Raman peaks of graphene (D peak at ~1350 cm^{-1}, G peak at ~1591 cm^{-1}, 2D peak at ~2700 cm^{-1}), the intensity ratio of the peaks (I_D/I_G, I_{2D}/I_G) was significantly different. In detail, the I_D/I_G ratio of DG-Catalyst graphene is much lower than that of the conventional CVD graphene (w/o Catalyst-B). This phenomenon was both observed under different growth temperatures (Fig. 2a for 1130 °C, Fig. 2b for 1170 °C), confirming the higher quality of graphene synthesized by the DG-CVD method. The conventional CVD graphene with and without gap have the similar I_D/I_G ratio, but the 2D peak intensity and the signal-to-noise ratio (SNR) of 'w/o Catalyst-F' is much smaller than 'w/o Catalyst-B'. This proved that the quality and growth rate of graphene was enhanced by a narrow gap. In especial, at the growth temperature of 1170 °C, DG-CVD graphene has I_D/I_G ratio of ~0.34 and I_{2D}/I_G ratio of ~1.47, indicating the direct growth of high-quality and single-layer graphene films on quartz substrates. In addition, the introduction of DG-Plate demonstrated the same catalytic effect on the growth of graphene on Si/SiO_2 substrates (Figs. S2 and S3), indicating the potential of DG-CVD method on various insulating substrates.

The homogeneity of graphene films was essential for further applications. Therefore, Raman mapping was demonstrated in Fig. 2c–f. The I_D/I_G value and I_{2D}/I_G value of two different approaches were compared, respectively. As for the "DG Catalyst" sample, 95% of the area has I_D/I_G ratios smaller than 0.4 and more than 91% of the area has I_{2D}/I_G ratios larger than 1.4, suggesting the formation of uniform and low-defect monolayer graphene. In contrast, "w/o Catalyst-B" sample has a non-uniform Raman mapping (Fig. 2c and e). 92% of the area has I_D/I_G ratios ranging from 0.8 to 1.1, and only 21% of the area has I_{2D}/I_G ratios larger than 1.4. This indicates a higher level of defects or amorphous carbon and a random layer number of graphene. Graphene films of "w/o

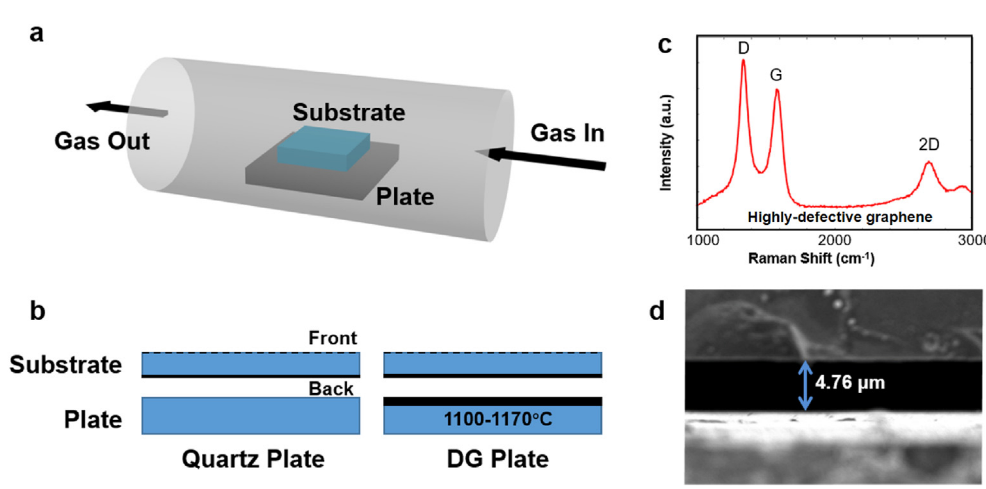

Fig. 1. (a) Experimental process of DG-CVD method; (b) Schematic of comparison experimental design; (c) Raman spectra of highly-defective graphene plate; (d) SEM image of the narrow gap between catalytic plate and substrate.

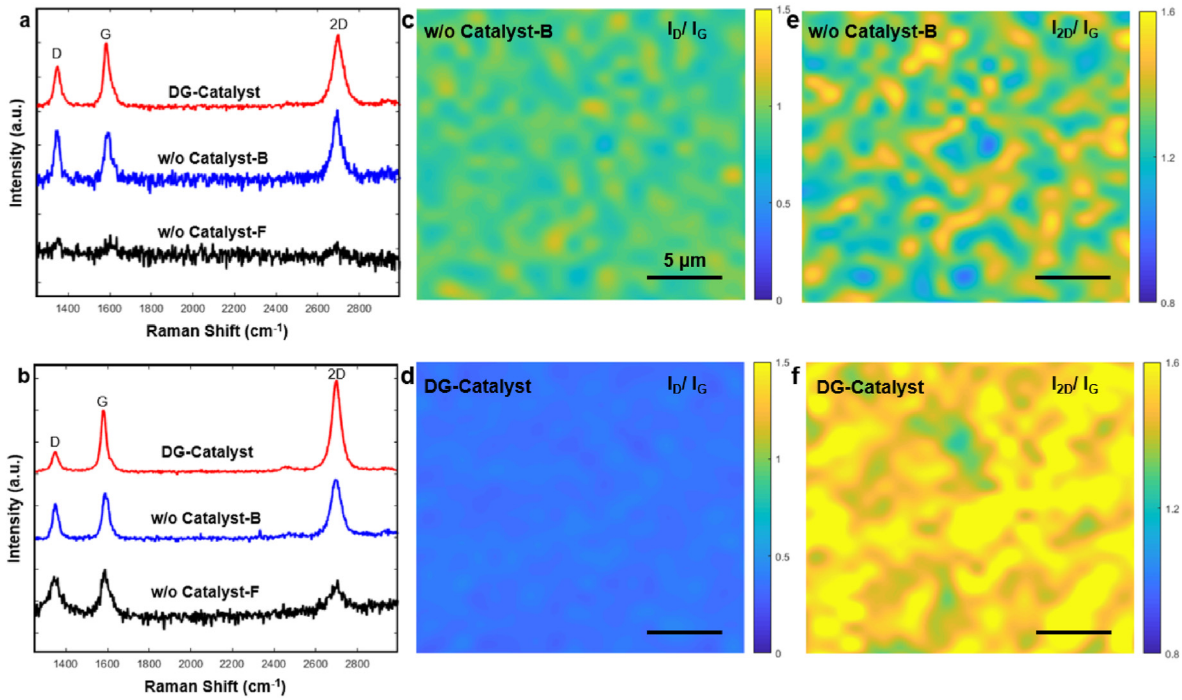

Fig. 2. Comparison of graphene films synthesized by DG-CVD and conventional CVD methods. (a–b) Raman spectra of graphene on quartz substrates, "DG-Catalyst" represents graphene glass synthesized through DG-CVD approach, "w/o Catalyst-F" and "w/o Catalyst-B" represent the front side and back side of conventional CVD sample, respectively, growth temperature is (a) 1030 °C and (b) 1170 °C. (c–d) Raman mapping of the intensity ratio of D peak and G peak of the "w/o Catalyst-B" sample and "DG-Catalyst" sample; (e-f) Raman mapping of the intensity ratio of 2D peak and G peak of the "w/o Catalyst-B" sample and "DG-Catalyst" sample.

Catalyst-F" sample was even more heterogeneous, longer growth time or higher temperature was needed without the catalyst of DG-Plate. The extremely slow growth rate of graphene on quartz by conventional CVD method was consistent with previous reports [15].

Fig. 3a–c demonstrate the evolution of graphene growth by DG-CVD approach ($Ar/H_2/CH_4$ = 200/45/2, 1130 °C). Graphene domains tend to grow isotropically on quartz substrates, which can be distinguished as round disks. With a 90 min growth, the coverage of graphene has already reached 79%, most graphene domains were connected to each other and formed a continuous film. Typically, only isolated graphene nuclei could be observed under this condition by conventional CVD approach. By increasing growth time to 120 min, the graphene film covered the quartz substrate almost completely (99%). Secondary nucleation was observed for longer growth time, which was marked in Fig. 3c (150 min). Apparently, the existence of DG-Plate and narrow gap greatly enhanced the decomposition of carbon precursors, thus significantly increased the nucleation time and the growth rate of graphene. As a matter of fact, fully-covered and high-quality graphene films were able to be synthesized in 120 min. In addition, there was a fascinating growth phenomenon observed under higher growth temperature (1170 °C). The nucleation was observed without the supply of carbon precursors (by setting the growth time to 0 min). As shown in Fig. 3d, the graphene nuclei have an average nucleus size of ~50 nm. It implies that, in a typical process, the nucleation was initiated before the growth stage (as long as the temperature was high enough). It is known that nucleation time was quite long for graphene growth on quartz substrates, which was probably caused by the high adsorption energy of methane [33]. In our experiment, the DG-Plate was also served as an additional carbon source. It facilitated the nucleation during the heating stage and thus shortened the growth time.

Fig. 3e and f demonstrate the remarkable differences in growth rate and graphene morphology under the same experimental conditions between the DG-CVD and conventional CVD approaches (1170 °C, 60 min). By comparing the coverage of graphene, there was no doubt that the growth rate was significantly higher by introducing a DG-Plate. As mentioned before, graphene domains were normally rounded on quartz [24]. However, different from other conditions, the morphology of graphene domains was more irregular in Fig. 3e. The reason causing this difference might be the higher concentration of active carbon species. With high temperature and catalyst of DG-Plate. The amount of active carbon species increased rapidly, graphene edges expanded so fast and led to different morphology of each domain. This phenomenon was also observed by a previous report (irregular graphene domains on Si/SiO_x was synthesized under 1185 °C) [22]. The detailed mechanism of graphene morphology on quartz needs further investigation.

The microscopic information of graphene synthesized by DG-CVD approach was further investigated. The graphene film is continuous and uniform as shown in the low-resolution transmission electron microscopy (TEM) (Fig. S5). Moreover, the layer number of graphene can be identified from the edge of sheet breakage from the high-resolution transmission electron microscopy (HRTEM) images [34]. As shown in Fig. 4a and b, DG-CVD graphene samples ($Ar/H_2/CH_4$ = 200/45/2 sccm, 2 h, 1130 °C) are primarily monolayer with limited multilayer regions. It is worth noting that the mixture of monolayer and few-layer graphene films is a feature of direct growth of graphene on insulating substrates, which is dominated by the higher possibility of multiple nucleation compared to the growth on metal substrates [35]. In addition, the C 1s X-ray photoelectron spectroscopy (XPS) spectrum was employed in order to clarify the bonding nature of carbon. The characteristic signals of graphene ($Ar/H_2/CH_4$ = 200/45/2 sccm, 2 h, 1130 °C) can be observed in Fig. 4c, which includes a sp [2] carbon peak (284.5 eV), a C–H peak (284.9 eV), and an extremely low and broad C–O peak (287.0 eV), indicating high quality of as-grown graphene

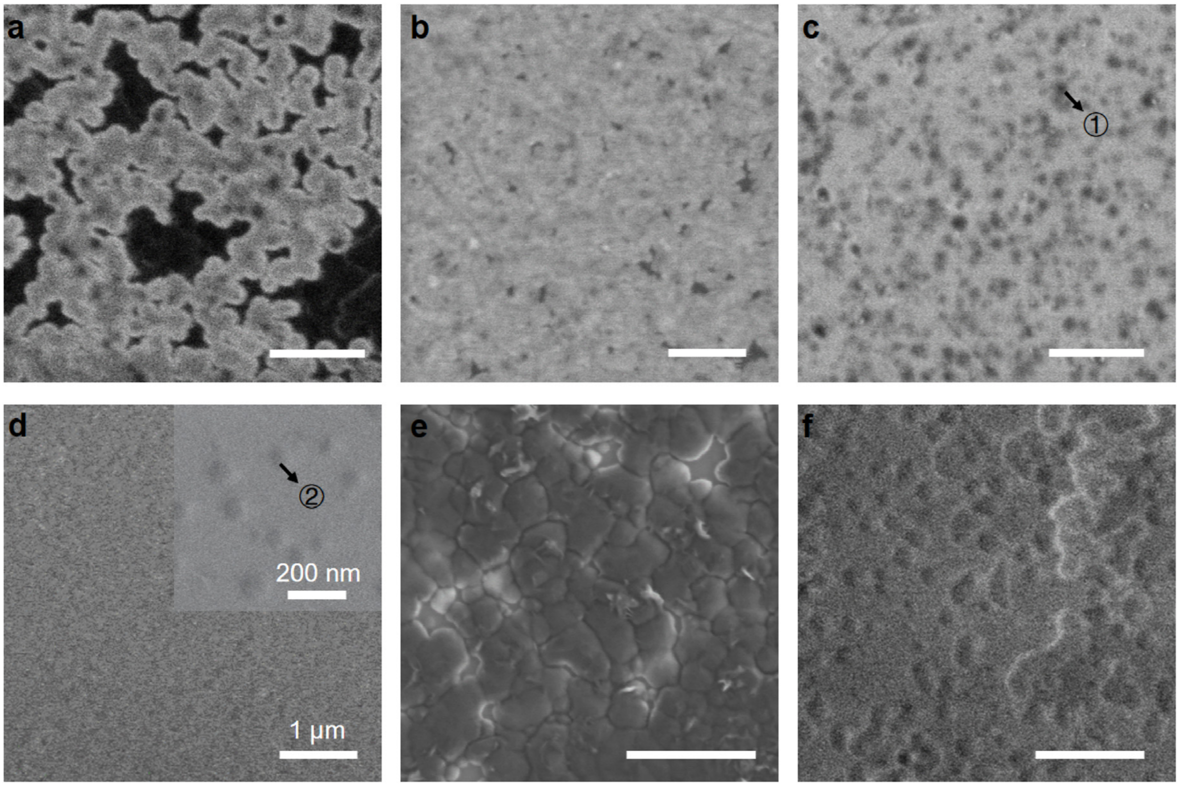

Fig. 3. Morphology characterization of graphene films grown on quartz substrates. (a–c) SEM images of graphene growth by DG-CVD method (1130 °C), the growth time is (a) 90 min, (b) 120 min, (c) 150 min; (d–e) SEM images of graphene growth by DG-CVD method (1170 °C), the growth time is (d) 0 min and (e) 60 min; (f) SEM image of graphene growth by conventional CVD method (1170 °C, 60 min).

on quartz by DG-CVD approach.

Fig. 4d displays representative optical images of the quartz substrates before (pristine quartz) and after DG-CVD process with different thickness of graphene films, which was tailored by varying the flow rate of H_2. The quality of these sample were investigated by Raman spectra as shown in Fig. S1. The optical transparency and sheet resistances of these samples were tested by UV–vis transmittance spectroscopy measurement and the four-probe method, shown in Fig. 4e. As the flow rate of H_2 decreased, the thickness of graphene films increased, leading to a lower transmittance and higher sheet resistance. The transmittance and the sheet resistance of the thinnest graphene film were 97.17% and 4.58 kΩ/sq, which was very close to the optimal values of mechanically exfoliated graphene (97.3%), the extra part of light absorption was caused by the limited multilayer regions. Specifically, the sample that we displayed in various former measurements ($Ar/H_2/CH_4$ = 200/45/2 sccm, 2 h, 1130 °C) has a transmittance of 87.75% and a sheet resistance of 3.12 kΩ/sq. Moreover, four feature points was selected to measure the sheet resistance as shown in the inset of Fig. 4f, demonstrating the uniformity of the electrical performance of the "DG-Catalyst" sample. On the contrary, the "w/o Catalyst-B" sample has an extremely uneven performance of sheet resistance, which was caused by the low coverage of graphene. Besides, the sheet resistance of "w/o Catalyst-F" sample was extremely high, most points exceeded the range of the four-probe device, and so was not given. Longer growth time was needed in order to achieve the practical applications for conventional CVD method.

Combining the optical transparency, electrical conductivity. The graphene glass fabricated by DG-CVD approach has the potential of application in transparent heating devices (Fig. S5c). As shown in Fig. 5a, with an input voltage of 25 V, the temperature of graphene heating device (0.7 cm × 1.0 cm, 3.12 kΩ/sq) reached ~43 °C in just 50 s. The inset of Fig. 5a presents a surface temperature mapping, demonstrating good uniformity of graphene glass. As the input voltage increased to 30 V, the maximum surface temperature was able to reach 70.4 °C, which indicated its practicality as a transparent heating device. Moreover, the contact angle of deionized water on pristine quartz and graphene glass with different transmittance was measured (Fig. 5b). The contact angle of pristine quartz is 73.94°, which is hydrophilic. On the contrary, all the graphene glasses have contact angle larger than 100°, indicating the switch of wettability before and after graphene growth. As the transmittance of graphene glass decreased, the contact angle increased slightly. For the thickest graphene glass in Fig. 4e (transmittance: 18.96%; contact angle: 105.07°), the contact angle is even smaller than the sample with the transmittance of 83.79% (contact angle: 105.33°). The thickness of graphene films has a very small influence on wettability, while the coverage of graphene was found to be more critical.

So far, graphene glass synthesized by DG-CVD method has been demonstrated in a series of experiments, representing excellent catalytic performance of DG-Catalyst. In order to understand the mechanism of catalytic behavior at the atomic level, density functional theory (DFT) calculations was further performed. Graphene growth on insulating substrates is a vapor-solid (VS) process, and the active carbon species in the gas phase dominate the graphene growth. Previous report has demonstrated that the CH_3 radical is critical for graphene VS growth on insulating substrates among possible active carbon species, which means that the growth rate of graphene is decided by the concentration of CH_3 radicals in the gas phase [36]. Therefore, we focused on the decomposition process

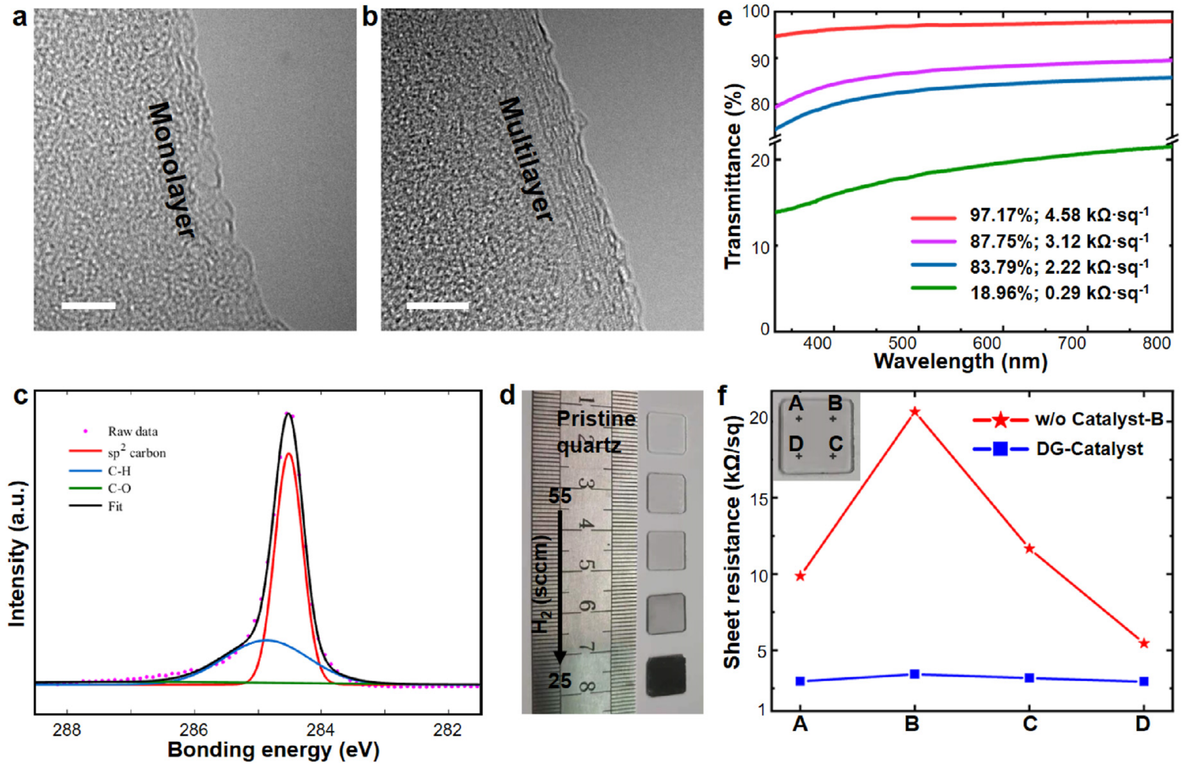

Fig. 4. Characterization of graphene glass. (a–b) HRTEM images of monolayer, the scale bar is 5 nm. (a) and few-layer (b) graphene; (c) C 1s XPS spectrum of graphene film grown on quartz glass, the characteristic signals of graphene with sp [2] carbon peak (284.5 eV), C–H peak (284.9 eV), and C–O peak (287.0 eV) were demonstrated in different colors; (d) Photograph of graphene glass before (topmost) and after DG-CVD process with different flow rates of H_2 (55, 45, 35, 25 from top to down); (e) Correlation of UV transmittance and sheet resistance of the graphene glass corresponding to the previous photograph; (f) Comparison of sheet resistance uniformity between "w/o Catalyst-B" sample and "DG-Catalyst" sample.

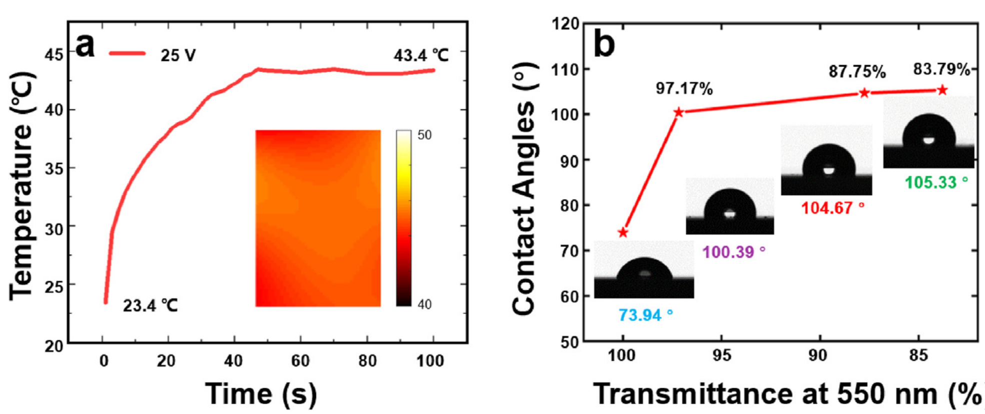

Fig. 5. Application of graphene glass synthesized by DG-CVD approach. (a) Surface temperature measurement of the transparent heating device with an input voltage of 25 V, the inset shows the temperature mapping of a 1 cm × 0.7 cm graphene glass; (b) Correlation of contact angle and transmittance of pristine quartz glass and graphene glass corresponding to Fig. 4e.

from CH_4 to CH_3. The growth process of graphene was not discussed in this study.

As confirmed by the D peak of Raman measurements, the DG-Catalyst has extremely high density of defects. According to previous reports, single vacancies, double vacancies, Stone-Wales defects, and zigzag/armchair edges are the most common defects of graphene. Specifically, we demonstrated the simulation of CH_4 decomposition on double vacancy, which is in the middle reactivity among all the common defects [37]. CH_4 decomposition without catalyst and with perfect graphene were performed for comparison. As shown in Fig. 6a, the calculated reaction barrier of CH_4 decomposition without any catalyst is as high as 4.74 eV, which results in a very slow growth rate of conventional CVD method. As for perfect graphene structure, the barrier of CH_4 decomposition is 4.08 eV, which indicates very low catalytic activity. In contrast, the reaction barrier of CH_4 decomposition on double vacancy defect is just 2.43 eV, showing important roles of graphene defects in decomposition of methane (More details in Fig. S6). Dangling bonds of these point defects might be the essence of lowing the reaction barrier [35,36]. Therefore, other defect structures are promising to be catalytically active as well.

Combining the results of experiments and simulations, the

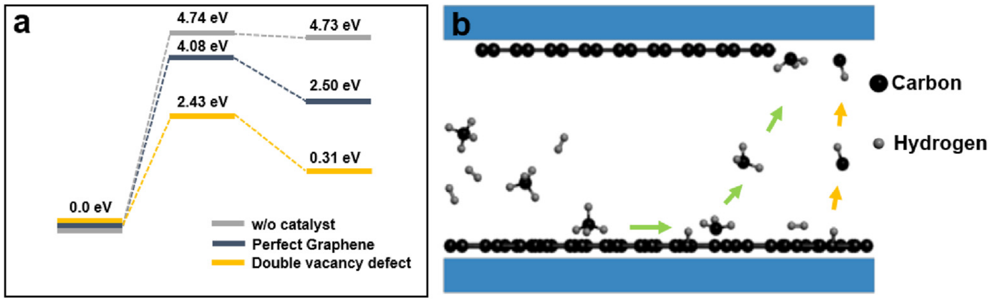

Fig. 6. Mechanism for the DG-CVD growth of graphene. (a) The energy profile for methane decomposition. To clearly show the catalytic role of graphene defects, reaction barrier of methane decomposition without catalyst, with perfect graphene structure and with double vacancy defects were showed for comparison; (b) Schematic diagrams of possible reaction routes for DG-Plate, including a catalyst process (green arrow) and an additional carbon supply process (yellow arrow).

mechanism of DG-CVD is further cleared. The schematic diagram was performed in Fig. 6b, which showed two important roles of highly-defective graphene in fast graphene CVD growth: first, the presence of graphene defects significantly lowers the barrier of methane molecules decomposition and so enhance the concentration of active carbon species (mainly CH_3 radicals); second, highly-defective graphene films have plenty of unstable regions, which can be etched by H_2, therefore become additional carbon source for graphene growth. The two important roles lead to an increase of nucleation rate and growth rate, ensure the fast growth of high quality graphene films on insulating substrates.

3. Conclusion

In summary, we have demonstrated a CVD method assisted by highly-defective graphene to achieve fast growth of high-quality and large-scale graphene films directly on quartz substrate. Through systematical experiments and simulations, highly-defective graphene was explored for two essential roles in the DG-CVD approach: (i) catalyst for methane decomposition; (ii) extra carbon source. In addition, a narrow gap was involved in this approach, which increased the collision probability and confined the active carbon radicals decomposed from DG-Catalyst. As a result, the growth rate and nucleation rate were significantly increased, single-layer graphene glass with high-quality was successfully fabricated within 2 h. Furthermore, the as-grown graphene glass was shown to be suitable for transparent heating devices in practical applications. The use of carbon materials as a catalyst for graphene synthesis is a hallmark of the DG-CVD approach, as it completely avoids the contamination of other elements. This method provides an attractive route to graphene production on insulating substrates for further practical applications.

4. Experimental section

4.1. The synthesis of catalyst plate

Prior to the growth of graphene, a highly-defective graphene glass was synthesized as the catalyst plate. In a typical experiment, quartz substrates were cleaned with deionized water, acetone and alcohol, and then dried under a nitrogen stream. The substrates were loaded into the commercially available tube furnace (AnHui BEQ Equipment Technology Co., Ltd) (Fig. S5b). The tube was pumped below 0.5 Pa to remove air, and then stopped pumping. The furnace was heated to the growth temperature (1100 °C) with 400 sccm Ar and 80 sccm H_2, the pressure inside the tube gradually approaches atmospheric pressure. After the temperature stabilized (15 min), the Ar/H_2 gas were converted to 200 sccm and 40 sccm respectively, and high flow of CH_4 gas (10 sccm) was fed into the reactor for 30 min. After the growth, the furnace was cooled down by sliding the heated zone away.

4.2. Growth of graphene

Graphene films were grown on quartz substrate by Atmospheric Pressure CVD (APCVD) with the assist of catalyst plates. The ultrasonically cleaned quartz substrate was placed above the catalyst plate at a distance of ~5 μm and loaded into the tube furnace together. After pumping below 0.5 Pa to remove air, the furnace was heated to the growth temperature and stabilized for about 15 min under a Ar and H_2 flow. Typically, the growth of graphene was performed at 1100 °C–1200 °C with a gas mixture of Ar (200 sccm), H_2 (25–55 sccm), and CH_4 (2 sccm) for about 1–3 h. After the growth, the sample was post-treated under a Ar and H_2 flow for 20 min and then cooled down by sliding the heated zone away. The schematic illustration of growth process was shown in Fig. S5a.

4.3. Transfer of graphene

Typically, the graphene film was transferred onto the copper grid for TEM characterization. The as-grown graphene glass was transferred with the aid of polymethyl methacrylate (PMMA) coating. The sample (Quartz/Graphene/PMMA) was immersed in an etching solution (Hydrofluoric acid/DI water/ethanol with concentration of 1:2:1) to dissolve the quartz substrates. Owing to the existence of PMMA, the sample was floated on the surface of the solution at first. Gradually, the graphene/PMMA film was detached from the quartz substrate. After the graphene/PMMA film was rinsed with DI water for 5 times, it was loaded onto a TEM copper grid and annealed at 100 °C for 15 min. Finally, the PMMA film was dissolved in acetone at 80 °C.

4.4. Characterization of graphene

The prepared samples were characterized sequentially by optical microscopy (Olympus BX51), Raman spectroscope (Thermo DXRxi, with laser excitation at 532 nm, and power of 7 mW), SEM (Hitachi S-4800, operating at 1 kV), TEM (FET Tecnai F20, operating at 200 kV), AFM (Bruker), XPS (AXIS Supra), UV–vis spectroscopy (Perkin-Elmer Lambda 950 spectrophotometer, detecting at 300 nm–800nm), four-probe resistance tester (Everbeing, DB-6), and contact angle tester (Biolin, THETA).

4.5. Fabrication and measurement of heating device

Cu wires were pasted on both edges of graphene glass with conductive silver glue to fabricate a heating device. The input voltages were applied to the defoggers by DC power supply

(HYELEC HY3005ET). Temperature of graphene glass was measured by K-type thermocouples and digital thermometer (SMART SENSOR AS877).

4.6. Calculations of methane decomposition

In the theoretical part of this work, decomposition of methane on various catalytic sites of highly-defective graphene was calculated. All calculations are implemented by Vienna Ab-initio Simulation Package(VASP) with Density functional theory [38,39]. The computer simulation cell consist of a 6x6 graphene supercell and a CH4 molecule. Periodicity boundary condition (PBC) is applied in the graphene plane (x-y plane) while a vacuum height of 22 Å perpendicular to the layer (z direction) to minimize the interlayer interaction. The edge atoms are fixed to avoid them running to the opposite boundary due to PBC and all other atoms are allowed to move during the relax process.

Projected augmented wave method (PAW) is adopted to compute the interactions between ion cores and valence electrons [40]. Exchange and correlation effects are described by generalized gradient approximation (GGA) with Perdew-Burke-Ernzerhof (PBE) functional [41,42]. The k-points in Brillouin Zone is set to $5 \times 5 \times 1$ with the Monkhorst-Pack mesh with Gamma centered [43]. DFT-D3 [DFT-D3 study of some Molecular Crystals] is used in all calculations to correct the weak van der Waals force. A plane-wave cutoff energy of 500eV was used and all the structures were fully relaxed with the energy and force convergence criteria of 10^{-6} eV and 10^{-2} eV/A. To help convergence, Gaussian smearing with 0.03 eV width was applied. The minimum energy paths and dissociation barriers of CH4 are calculated with climbing image elastic band (CI-NEB) method [44].

CRediT authorship contribution statement

Yue Li: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Visualization. **Yong Liu:** Investigation, Validation, Writing – review & editing. **Yunbiao Zhao:** Investigation, Validation, Writing – review & editing. **Yifan Zhang:** Validation. **Yi Chen:** Validation. **Qining Wang:** Supervision, Funding acquisition. **Ziqiang Zhao:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.carbon.2021.11.019.

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